

NDNSF: A DATA-CENTRIC SERVICE FRAMEWORK FOR SECURE, COLLABORATIVE,
AND DYNAMIC NETWORKED APPLICATIONS

by

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Abstract

Supporting distributed services over Named Data Networking (NDN) requires discovering willing Providers at runtime, authorizing service invocation, and coordinating dependent computations through named data. This proposal investigates how NDNSF can integrate these functions while making their security guarantees and performance trade-offs explicit. NDNSF uses attribute-based encryption (ABE) to encrypt discovery descriptors for authorized Providers without requiring the User to enumerate their individual identities. For service authorization, NDNSF combines verification of responses to encrypted challenges with signature validation under trust rules and request-state checks. Service permissions are managed independently of identity-signing credentials. Authorized but unselected Providers can still read the descriptor; after selection, request-scoped keys protect application inputs and results.

The proposed collaboration extension assigns roles using eligible positive acknowledgment (ACK) Data and derives Provider-local rules for accepting named dependencies. These rules are intended to restrict intermediate inputs to Data from assigned producer roles over declared dependency edges, whether retrieved from a producer or a cache. Multi-view collection and recognition by unmanned aerial vehicles (UAVs), together with distributed inference, provide application settings for evaluating this approach.

Existing prototype studies provide preliminary evidence on service invocation, mobility handling, and execution efficiency under specific configurations. They motivate, but do not establish, the effectiveness of the proposed collaboration mechanism. Planned evaluation will compare complete authorization alternatives, test role and dependency enforcement during normal execution and failure recovery, and measure communication and processing overhead. The research aims to establish when data-centric discovery, authorization, and collaboration improve distributed service execution, and when their costs outweigh the benefits.

Keywords: Named Data Networking, service authorization, provider discovery, collaboration, distributed inference

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Chapter 1

Introduction

1.1 A Task That Requires Runtime Decisions

Consider an operator requesting several views of a ground location. Nearby UAVs may have different viewing directions, sensors, battery levels, and availability. A compute-capable UAV or edge node can analyze the collected images. The operator knows the required services but cannot assume which Providers will be reachable and willing to participate when the request is issued. The application must discover candidates, assign distinct collection and processing roles, and associate each intermediate image with its assigned producer and consumer. This is the motivating workload, not a claim that multi-view recognition itself is new.

A discovery descriptor contains the information needed to decide whether to offer service: for example, a sufficiently precise area, deadline, sensor requirement, and processing requirement. Application input contains the material needed to execute the selected task. The boundary is semantic, not a fixed field list: a location needed for feasibility belongs in the descriptor, whereas an image needed only for processing can be provided after Selection. Disclosure of an area and deadline can reveal an operator's intended activity even without revealing the images. We therefore study discovery that keeps descriptor Content confidential from parties lacking the required decryption capability. Authorized candidates, including those not selected, can read it; names and traffic patterns are not thereby hidden.

1.2 What NDN Already Provides

Named Data Networking (NDN) provides Interest-based retrieval of named Data, with signatures that support integrity and producer authentication under a trust policy [1, 2]. Caching can satisfy a later Interest without contacting the original producer; it does not guarantee persistent storage or delivery before a deadline. A signature alone does not establish execution permission, freshness for a particular task, or semantic correctness of a result. NDN Sync

announces changes in distributed datasets so applications can retrieve newly available Data [3,4]; it is not an unsolicited Data-push primitive.

NDN service and computation frameworks already exist, including NFN, NFaaS, RICE, and NSC [5–8]. NDNSF is not motivated by their absence. The research concerns the integration and evaluation of runtime candidate discovery, service authorization, and task-specific dependency enforcement. A broad claim that host-oriented systems cannot support these functions would also be incorrect: for example, gRPC supports extensible name resolution [9]. Comparisons must examine complete configurations and their costs.

1.3 Thesis and Research Questions

The thesis to be evaluated is that a service framework can use named, verifiable objects to connect runtime Provider selection to explicit execution rules, while separating service-level discovery confidentiality from request-specific input and result confidentiality. The expected benefit is conditional: additional discovery, validation, and key-management work must be justified by the task’s mobility, authorization, or collaboration requirements.

- RQ1. How can NDNSF discover willing Providers while keeping discovery descriptors confidential from unauthorized readers and checking service permissions before execution, and at what cost relative to complete alternative designs?
- RQ2. How can Providers selected at runtime execute a task according to its roles and named dependencies, whether intermediate Data is retrieved from producers or caches?
- RQ3. Under what workload and network conditions do these mechanisms improve completion, latency, or execution efficiency, and when do their costs outweigh their benefits?

UAV sensing and distributed inference are complementary validation workloads for these questions, not separate research contributions simply because two applications are implemented. Recovery is part of the protocol and evaluation scope; it is not assumed to be an independent novel contribution.

1.4 Contribution and Evidence Boundary

The intended contributions are (1) a justified and evaluated discovery/authorization design, (2) a protocol that converts runtime role assignments into enforceable named-object admission rules, and (3) bounded empirical findings about their benefits and costs. ABE, Trust Schema, Sync, caching, model partitioning, and challenge-response are reused mechanisms, not newly invented primitives. The work claimed here concerns their NDNSF-specific design and evaluation. The final dissertation must identify the candidate's implementation, analysis, and experiments separately from collaborators' work and reused software; this proposal does not assign unverified personal ownership to shared code.

This is a revised proposal, not a completed dissertation. The submitted v0.1 implementation, subsequent measured campaigns, current implementation work, and proposed acceptance conditions are separate evidence categories. Existing request-ID checks associate validated messages with active requests; the identifier alone neither authenticates a message nor grants permission. A hypothetical stale recovery attempt is not a defect established in a single-attempt historical protocol. Current collaboration and native distributed-inference work must pass its own end-to-end gates. Chapter 5 reports only the limited findings presently available.

Chapter 2

Related Work and Design Alternatives

2.1 Invocation Is Not the Unsolved Problem by Itself

NFN studies named computation and expression evaluation; NFaaS studies execution placement in a named-function environment [5, 6]. RICE provides an invocation handshake with support for authentication and security information [7]. NSC specifies signed invocation Interests validated using an application-provided Trust Schema, and discusses protected input and result exchange [8]. These are positive precedents for invocation and authorization, not insecure straw-man baselines. A workload-specific implementation of NSC used in an experiment is not the entire NSC design.

For NDNSF, the narrower question is how a service request gathers current willingness and feasibility information before selecting execution roles, and how the resulting assignment constrains intermediate-object use. Long-term permission to offer a service does not by itself determine which authorized Provider is assigned to a particular predecessor role. Neither a request ID nor a valid signature alone expresses that assignment. Conversely, role and edge checks are not automatically novel merely because their fields are placed in a manifest. A final novelty claim requires a feature-level comparison with service composition, workflow security, and capability-based systems, including CaSCON [10], rather than a claim of being the first NDN service framework.

2.2 Comparing Authorization Designs under the Same Requirements

DNMP is a concrete example of role-constrained command signing and verification [11]. Role granularity is policy-defined: a role can cover multiple actions, and credentials can be reused where the rules permit.

Our comparison supplements DNMP-inspired authorization with request-state checks and content encryption. These are requirements for an NDNSF alternative, not claims that the

Table 1: Alternatives under the same discovery and execution requirements.

Design	Execution decision	Non-public discovery	Management cost
DNMP-inspired roles	Validate signed commands against role and command rules	Add encryption and a group/key-distribution design; signatures alone do not conceal Content	Role credentials, trust rules, status or expiry checks, and encryption keys
NDNSF ABE-backed	Check signatures under trust rules, challenge responses, and request/authorization state	Encrypt a descriptor for a service-authorized receiver set without listing current candidates	Identity credentials, ABE policies/DKEYs, status distribution, and rekeying

original DNMP implements the NDNSF message workflow.

Recipient-specific encryption protects selected exchanges. Confidential discovery additionally needs a way to distribute decryption capability before current participants are selected, such as group-key management or ABE. These are alternatives, not an exhaustive list. Trust Schema alone does not provide confidentiality, but a system combining it with suitable encryption can.

For revocation, a credential-based alternative must reject a revoked credential using sufficiently current status information or wait for expiry. Removing a certificate from a repository cannot invalidate copies already held by a validator. This is a proposed lifecycle requirement for a fair alternative, not a claim that the original DNMP paper implements NDNSF’s runtime revocation protocol.

2.3 From a Confidentiality Requirement to an Architectural Benefit

NDNSF’s two connected design goals are confidential discovery before Provider selection and separation of dissemination from fine-grained content access. To meet them, the User encrypts descriptors and publishes the ciphertext for shared retrieval, while the Controller issues policy keys with different access scopes. NAC-ABE supplies attribute-based control

of Data decryption; the enhanced library includes KP-ABE support [12]. NDNFS reuses this permission-management mechanism for invocation: participants verify protected challenge responses, message signatures under trust rules, and current authorization/request state. These checks are separate from content encryption. Withdrawal and rekeying are lifecycle requirements with costs, not independent evidence of superiority. Section 3.3.1 develops these design choices and their limits.

The comparison must include group-key or broadcast-encryption alternatives where appropriate. A stable, small group can make conventional group encryption attractive; known Providers and simple permissions can favor signed invocation with recipient encryption. ABE becomes attractive when recipients are identified by service entitlement before their current identities are known, and multiple entitlements should remain independent of identity-signing credentials. Policy size, membership churn, encryption cost, and authority availability bound this argument.

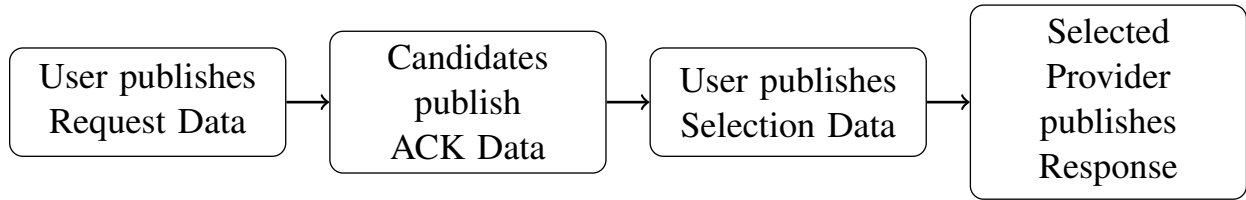
Chapter 3

NDNSF System Model and Protocol

3.1 Participants, Trust, and Exclusions

These are protocol roles, not necessarily separate devices: one node may be both a User and a Provider. The managed organization registers identities and approves service permissions; the Controller administers that policy. Participants require a provisioned trust anchor and validation rules for identity certificates and Controller material. Not knowing which candidates will respond to this invocation does not mean that the authorization authority has never enrolled those entities. Billing, arbitrary business transactions, globally optimal scheduling, and protection against all denial-of-service traffic are outside the claimed scope. The User requests and coordinates an invocation. Providers offer and execute services. A Controller administers authorization material and signed status. Forwarders and caches can deliver Data without being trusted to authorize execution. The model trusts the Controller and the User's planning decision; it does not claim consensus against a malicious coordinator. An attacker may observe visible names and traffic, inject invalid objects, replay previously valid messages, or operate without the needed service permission. Credential compromise and intentional sharing by an authorized principal require additional controls and are not prevented by ABE alone.

Authorization is permission to perform an operation, authentication establishes the signing identity under trust rules, and confidentiality limits who can recover Content. Valid credentials do not prove honest load reports, correct inference, or truthful sensor observations. Signed status must be obtained and validated before it can affect local authorization; a disconnected participant cannot be assumed to learn revocation immediately. Deadline/expiry checks require an explicit clock policy. Tests must report the allowed skew and failure behavior rather than silently assume synchronized clocks.



Controller: authorization material and signed status.
 Sync: discover names. Interests: retrieve Data.
 User/Provider: validate Data, check permissions
 and request state before acceptance.

Figure 1: Logical invocation dependencies. Arrows express application causality, not unsolicited network Data delivery.

Table 2: Logical message responsibilities; this is not a wire-format listing.

Data	Purpose	Acceptance condition
Request	Publish service requirements, minimized discovery descriptor, request identity, and Requester Challenge	Validate signature and current authorization context; recover the protected descriptor before evaluating willingness
ACK	Report positive or negative participation decision; a positive ACK carries selection information and Provider Challenge	User checks signer under trust rules, service permissions, request ID and Requester Challenge; only eligible positive ACKs enter planning
Selection	Assign the chosen Provider/role and supply protected input or input references and request-scoped key material	Provider checks the User's signature under trust rules, intended Provider, challenge response, request state, service permissions and applicable plan rules
Response	Publish completion/result metadata or references to protected results	Check authorized producer, request/plan context, result identity, and decryption scope

3.2 Four Named Message Types

These are NDNFSF Data objects. Sync supports discovery of their names, after which Interests retrieve Data, possibly from a cache. A negative ACK is an application participation decision, not a network Nack. Not every observed Request must receive a reply: invalid, unauthorized, duplicate, or expired messages are subject to validation and state checks. A positive ACK expresses willingness; it does not itself start the selected task or guarantee later resource availability.

3.3 Authorization and Confidentiality

3.3.1 *Rationale for ABE-Backed Service Authorization*

NDNSF's rationale has two connected parts: a security requirement for confidential discovery before Provider selection, and an architectural benefit of separating dissemination from fine-grained access to protected content. The first explains what must remain confidential; the second explains how different access scopes can coexist with shared dissemination. These are not two independent proofs of ABE's superiority. Permission aggregation and identity-credential reuse support the second part; permission withdrawal is a lifecycle requirement with costs, not a third reason to prefer ABE. Execution authorization determines whether a User may invoke a service; Provider authorization determines whether an entity may offer that service. Content confidentiality limits who can read protected content. The Controller administers these service permissions; the User and Provider check them during invocation. Signature validation under trust rules and invocation-state checks remain necessary. Permission to offer a service is distinct from selection for one particular task.

Reason 1: confidential discovery before Provider selection. Before Provider selection, the User publishes a discovery descriptor containing the location, deadline, and resource requirements needed for a Provider's positive-ACK decision. For example, nearby UAVs need an observation area and arrival deadline to decide whether to participate, but do not yet need the

full application input. The User encrypts this descriptor using ABE and the requested service's attribute, without enumerating Provider identities. Confidentiality is against parties lacking the required decryption capability: authorized but unselected Providers can still read the descriptor. Content encryption does not hide outer Data names or traffic patterns. In the request-scoped confidentiality path, the full application input is encrypted with a request-scoped key; Selection protects that key for the selected Provider. The requesting User retains access. A Trust Schema alone does not encrypt content; a role-authorized design can also use ABE or service-group encryption to provide discovery confidentiality.

Reason 2: separating dissemination from fine-grained access control. The set of nodes that can retrieve encrypted Request Data need not be the set that can read its discovery descriptor. Forwarders, caches, and repositories can carry or retain ciphertext without receiving its decryption keys. The User can publish according to service attributes rather than construct a recipient identity list for each Request; the Controller still authorizes entities and issues their keys. This separates delivery from disclosure, rather than delegating plaintext filtering to each distribution node.

Different entities can hold overlapping service entitlements while using the same dissemination mechanism. A UAV application may, for example, authorize one User for observation and detection services and another for detection and mapping services. This is an illustrative service-permission distribution, not evidence that arbitrary task, location, or time predicates are implemented. Fine-grained access survives aggregation only when the relevant objects retain their separate cryptographic scopes: encrypting a combined plaintext under one condition exposes the entire plaintext to anyone satisfying that condition.

In KP-ABE, ciphertext is associated with attributes and a decryption key (DKEY) encodes an access policy [12]. NDNSF associates Provider access with `/SERVICE/<service>` and User access with `/PERMISSION/<service>`. Within one authority and ABE parameter generation, it combines an entity's service entitlements into an

OR policy for its individually issued DKEY, while reusing its identity-signing credentials. Trust Schema specifies permitted relationships between Data names and signing key names; a validator applies those rules and checks the chain to a trust anchor. The schema by itself does not restrict who can read copied Content [13]. This distinction applies beyond role-based designs. Trust Schema can express fine-grained rules and can be combined with encryption. NAC also protects distributed content through encryption and key distribution [14]; ABE's distinctive contribution here is attribute-policy-based decryption, not an exclusive ability to protect cached data or reuse credentials.

DNMP constrains command targets and actions through role signing keys and a Trust Schema [11]. Existing roles may cover several permissions; different entitlements may require changes to role assignments, credentials, or rules, not necessarily one new role per service.

The comparison therefore considers DNMP-inspired signing authorization with separate content encryption. ABE separates service-permission policies from identity-signing credentials; DNMP-inspired designs express permissions through role assignments and trust rules.

Neither design has a fixed credential cost. ABE requires DKEY issuance and updates, with costs that can grow with policy complexity. Role-based designs can reuse credentials; credential counts alone do not establish lower total cost.

3.3.2 Authorization Mechanisms, Lifecycle, and Costs

Connecting decryption capability to invocation authorization. In the ABE-backed invocation path, the User includes a Requester Challenge (`UserToken`) in protected Request content. A Provider that recovers it returns it in ACK Data; the User checks it against the active request. The Provider places its own Challenge (`ProviderToken`) in protected ACK content. The Provider checks the corresponding challenge response in Selection before starting the selected task.

These checks demonstrate access to protected challenge values, not the creation of new permissions or a formal proof of exclusive key possession. The User and Provider also validate

message signatures and certificate chains under trust rules, check current service permissions, and match the request and its state. One-time state prevents a repeated message from authorizing another execution only while that state is preserved or safely invalidated on restart. Authorized key or challenge sharing is outside the claimed threat protection. This reuses attribute-based permission management rather than a separate set of role-based invocation rules. It is not cost-free: confidential discovery already needs Provider DKEYs, whereas User permission DKEYs and challenge processing are additional authorization costs. ABE confidentiality combined with DNMP-inspired role authorization remains a valid alternative.

Runtime permission withdrawal and evaluation. The current runtime implements permission withdrawal through Controller-signed authorization status and checks at invocation processing points. A participant must retrieve and validate updated status before enforcing the newly withdrawn authority. ABE parameter rotation and refreshed DKEYs protect subsequent exchanges encrypted under the new generation. The present rotation is Controller-wide, so other authorized members may also need refreshed DKEYs; it is not a zero-fan-out revocation scheme.

DNMP describes encryption-key rotation to restrict subsequent access to measurement data, but does not detail a comparable online command-authorization revocation protocol [11]. A DNMP-inspired comparison design can revoke the relevant credential and require execution nodes to check signed revocation status or reject expired certificates. This invalidates authority at the validator; it does not physically retrieve distributed certificates. Both approaches require update propagation, and neither retracts previously disclosed keys or plaintext.

Focused integration tests exercise NDNSF's runtime revocation checks, including identity-, certificate-, and service-scoped withdrawal. These tests establish behavior at the exercised runtime boundaries, not network-wide instantaneous revocation or lower cost than certificate-based alternatives. The remaining evaluation should measure status-propagation delay, rejection delay, rekeying traffic, and effects on unaffected members. Runtime status validation and key rotation therefore extend NDNSF to permission withdrawal, with explicit propagation

and rekeying costs.

Discovery confidentiality applies to descriptor Content against parties lacking the required ABE decryption capability. It does not conceal the descriptor from other authorized candidates or provide traffic-analysis resistance. After Selection, the design uses request-scoped symmetric keys for application input and results; the selected Provider receives the necessary key material protected for its authenticated public key. Encrypting an image symmetrically and delivering its key in Selection is distinct from encrypting the image directly with a public key. The response key is shared by the requesting User and the selected Provider: other Users are excluded under the key-secrecy assumptions, but the Provider necessarily knows the plaintext it produces. This paragraph specifies the intended protected exchange, not retroactive behavior of every v0.1 path.

3.4 From ACKs to a Collaboration Plan

The application supplies a role/dependency template, required services, feasibility constraints, and deadline. The template may be written manually; automatic generation is not necessary to establish the framework contribution. A replaceable planning policy selects among eligible positive ACKs after the collection window closes. Closing the window freezes the candidates observed for that decision, not all possible Providers in the network. ACK claims can become stale, so a selected Provider still performs execution-time admission and may reject work it can no longer execute.

In the proposed design, the User signs a Plan Manifest containing the request identity, accepted candidate snapshot, exact role-to-Provider assignments, permitted named dependency edges, authorization/key scope, and execution identity. Its digest is computed over a canonical plan body that excludes the digest field itself. Providers validate that signed plan before deriving local dependency checks; a long-term certificate does not specify this task's assignments. The User must record the committed plan and reject a different plan for the same execution identity. This is a local state-machine rule under the trusted-User model, not a consequence of signing alone or protection against a Byzantine User signing two plans.

3.5 Object Admission and Execution State

Under the proposed acceptance rules, a downstream Provider accepts a dependency only if its Data signature validates under the applicable trust rules, its request/plan identity matches the active assignment, the signer is assigned to the predecessor role, and its name, data format, and key scope match a declared edge. Required segments must validate, and incomplete or malformed objects cannot satisfy an edge. The same checks apply to Data fetched from a producer, cache, or repository. Names belong to the actual producer’s authorized namespace; the plan references those names rather than relabeling another producer’s object as the consumer’s own Data.

Table 3: Proposed task-state contract and required negative checks.

State/event	Permitted transition	Required rejection or bounded behavior
Collecting ACKs	Close window and validate candidate set	Late ACK does not silently alter a committed plan
Selected	Validate Selection and perform local admission	Positive ACK alone, wrong Provider, or spent challenge cannot start work
Waiting for dependencies	Execute when required authorized edges are satisfied	Wrong role, undeclared edge, incomplete object, or wrong active execution is rejected
Completed/cancelled	Return recorded status under duplicate-message policy	Duplicate publication does not implicitly repeat a side effect
Failure/reassignment	Explicitly terminate or authorize a replacement execution	Old execution must not acquire authority from the new assignment; allowed result reuse is explicit

These are proposed acceptance conditions, not a claim that every production path is qualified. Network re-retrieval is not identical to repeating computation. If a task is reassigned, cancellation, fencing, and result-reuse rules must be explicit. An already computed image may remain useful when the new plan explicitly authorizes its reuse; an old object is not automatically invalid solely because it is cached. Application-defined compensation can limit side effects but does not provide distributed atomic transactions or exactly-once physical action.

The authorization design and comparative evaluation are intended to answer RQ1; the proposed plan/state rules and their tests are intended to answer RQ2. Neither question is answered by a design description alone. Existing invocation evidence does not establish collaboration correctness; wrong-role, wrong-edge, replay, cancellation, and direct/cache tests remain separate evaluation obligations.

3.6 Conditional Correctness Argument

Assume uncompromised signing keys, correctly enforced trust/authorization checks, and Providers that obey the specified state machine. Selection-before-execution follows only if every handler-start path checks an active assignment and unspent authorization state. Dependency integrity follows only if every input path, including a cache hit, checks the assigned producer and declared edge before satisfying the dependency. Replacement isolation additionally requires old workers and late callbacks to lose authority under the new execution identity. Restart recovery must preserve or safely invalidate the relevant one-time state; signature validity and cache FreshnessPeriod alone are insufficient.

These are conditional invariants and implementation proof obligations, not a cryptographic proof of a complete system. Tests should inject a wrong-role object within the same request, an undeclared edge from a correctly signed producer, and a late callback after replacement. Each isolates a property that a request ID alone cannot express. Application result correctness and physical side effects remain outside this argument.

Chapter 4

Application Validation

4.1 UAV: Multi-View Collection, Processing, and Return

The primary UAV case is a ground-location request, not a complete autonomous mission system. An operator chooses a target location or area on a map. The Request reveals only the service-level-protected information needed to assess coverage, viewing feasibility, sensor capability, and timing. Eligible UAVs return positive ACKs; a planner assigns complementary viewing roles and a processing role. Selection delivers the execution details and appropriate keys to the chosen Providers.

In the proposed UAV workflow, each sensing UAV signs ViewImage Data containing encrypted image content and publishes it in its own authorized namespace, with observation time and available pose/calibration metadata. Content encryption does not conceal the outer Data name. GPS helps relate platforms and target areas, but does not by itself establish line of sight, camera direction, accurate depth, or correspondence between views. The application must define capture tolerances, view adequacy, and failure handling. Different views cannot be treated as identical cropped inputs without matching or geometric reasoning.

The compute Provider retrieves the authorized images by their named dependencies, performs detection or other specified analysis, and publishes signed annotation Data. Its result references the original image Data and includes the permitted image keys in Content encrypted for the requesting User. This reuses the original signed images; it does not republish them under a false producer identity. The User retrieves those images and annotations and renders the overlay. Improved recognition or localization accuracy is not assumed: such a claim requires ground truth, a single-view control, and a specified fusion method.

MiniNDN [15] tests named-data execution under controlled connectivity; PX4 SITL [16] can test the application/autopilot interaction. Neither is a real-flight safety certification. The framework-level evaluation objective is authorized multi-view collection, processing, and return

before a deadline. Other flight-control, video, recording, and patrol features remain supporting engineering, not independent dissertation completion conditions.

4.2 Distributed Inference: Explicit Data and State Dependencies

The second workload partitions a specified model across Providers and exchanges named intermediate objects. Pipeline partitioning assigns sequential layers or layer groups; tensor partitioning divides operations/tensors and requires the appropriate communication and collectives. The terms describe different execution structures, not interchangeable drawing orientations. ONNX [17] represents computation graphs, but export alone does not implement cross-Provider collectives or prove distributed numerical correctness. Model partitioning has substantial prior work [18, 19]; the proposed NDNSF increment is runtime assignment and checked named dependency execution, not partitioning itself.

In the proposed execution, the planner combines a supported model/partition description with ACK capabilities. Providers obtain authorized artifact references, prepare their assigned fragments, and exchange activations or collective inputs over the declared edges. A repository provides managed artifact storage; the forwarding Content Store is opportunistic caching, not a persistence guarantee. Reuse must distinguish an identical model artifact from request-specific activation or execution state.

For autoregressive generation, the model computes K/V tensors; the runtime retains and supplies the relevant cache state to later decoding steps. Keeping model weights resident does not eliminate this state dependency. Pipeline execution still has a temporal feedback loop: generated-token/state information is needed for the next step. If LLM streaming is retained in the final claim, evaluation must separate prefill, time to first token, per-token decoding, and KV storage/transfer. It is not necessary to qualify every model or hybrid partition merely to validate RQ2.

4.3 Current Scope and Completion Gates

Current native implementation work and local tests are evidence of engineering progress, not interchangeable with cross-process, multi-device, or sealed-container qualification. As of the September 2026 revision, the V3/native integration and compatibility gates are not all closed. The historical small-model campaign in Chapter 5 does not qualify canonical-layer reuse, tensor/hybrid execution, multi-accelerator placement, or recovery of a large language model.

The final campaign will freeze model version, precision, supported partition, device memory, network conditions, warm/cold state, and an output-comparison tolerance. It will include a single-Provider baseline when feasible. If the model cannot fit one device, distributed feasibility is meaningful, but no speedup ratio against a non-running baseline is claimed. UAV and DI jointly test the role/dependency abstraction; their task algorithms and device-specific adapters remain application responsibilities.

Chapter 5

Available Evidence and Required Evaluation

5.1 Measured Findings Are Configuration-Specific

The following numbers are retained from recorded campaigns, not new experiments in this revision [20]. The independent unit for comparative uncertainty is the seed/run, not each correlated request. Success-conditioned latency must be read with completion and timeout rates. A baseline label identifies the evaluated harness configuration, not every feature of the framework named in that label.

Table 4: Selected evidence and limits; values are unchanged from their records.

Campaign	Observation	Supported interpretation / limit
Ten-seed switching holdout	1,312 matched switching requests; NDNSF minus gRPC seed-p95 difference: -597 ms, 95% CI $[-992, -200]$; versus NSC: -2618 ms, CI $[-2916, -2315]$	Lower conditional tail latency in the tested four-Provider configuration, not universal mobility or success-rate superiority
Six-seed work-efficiency repeat	NDNSF: 1,798/1,800 completed, 1,800 execution starts; gRPC-PAR-4: 1,800/1,800, 7,180 starts	Selection avoids redundant starts relative to this parallel policy; starts are not messages, bytes, CPU time, or energy
Earlier small-model DI diagnostic	Ten successful runs/layout: mean 259.4 ms distributed versus 186.8 ms single Provider; p95 283.9 versus 206.3 ms	Splitting was slower for this model/configuration; small success-conditioned sample does not establish a general tail-latency result

In the work-efficiency comparison, NSC-4 also completed 1,800/1,800 requests, with 1.002 execution starts per request (rounded). This does not establish a work-efficiency advantage over NSC. The mobility campaign uses common traces, publication times, deadlines, and four Providers per system. The switching subset is defined by reachability and the registered classification, not chosen because NDNSF wins. Overall coverage controls and excluded/failed requests must remain alongside this conditional result in the final analysis. The reported small-model difference is consistent with communication/setup overhead, but no isolated

profiling result here establishes its cause.

The transition experiment reports 357/357 completions for NDNSF without pre-registering Provider addresses, 0/357 for static-list baselines, and 357/357 for pre-registered baseline variants. It isolates the discovery configuration. A gRPC resolver or other discovery integration can update a candidate list [9]; the result does not prove that gRPC or NSC cannot discover a new Provider.

Historical v0.1 loss results include 599/600 completions at 1% per-link loss, 598/600 at 3%, and a best recorded 539/600 at 10%. The last is a selected historical run, not a current aggregate estimate or a matched comparison against later baseline data. The expression $1 - (1 - \ell)^4$ assumes four independent packet traversals with equal loss ℓ ; it is not the full service-failure probability after retries. Timeouts under heavy loss do not, without further interventions, identify one exclusive cause or rule out protocol/policy effects.

5.2 Evidence Needed for Each Research Question

Table 5: Claim-driven evaluation plan, not a completed-results checklist.

RQ	Comparison	Correctness / failure observation	Costs and outcomes
RQ1	ABE-backed authorization; role rules plus content encryption; suitable group-key and authorization combination	Unauthorized call never starts handler; replay, wrong challenge, revoked/expired state, status delay	Completion, discovery delay, crypto work, credential/policy size, renewal and withdrawal traffic
RQ2	Same task graph with static/dynamic assignment; controlled removal of each admission check	Wrong role, edge, name, key or execution; direct/cache parity; incomplete objects; cancellation or reassignment	End-to-end completion, validation cost, intermediate bytes, redundant work, bounded recovery
RQ3	Matched Provider counts, tasks, capacities, deadlines and failure schedules; feasible single-device control	Include failed and timed-out requests; distinguish network, admission, execution and result-retrieval failures	Completion and latency together; cold/warm costs; per-run uncertainty; conditions with no benefit

Each campaign must preserve its source/dependency/model identity, configurations,

seeds, measurement start/end points, raw-run references, and analysis scripts. Scaling dimensions should follow the actual claim: candidates and services for discovery/authorization, edges and object sizes for collaboration. Ablation diagnoses a mechanism; an intentionally insecure variant is not a recommended deployment. A startup or collector failure must be diagnosed before attributing it to the harness or protocol. Preserve failed attempts; do not count an experiment that never reached its measurement boundary as a measured protocol failure rate.

Chapter 6

Research Plan, Risks, and Decision Gates

6.1 Scope Before More Features

The immediate priority is a frozen, runnable baseline for the final claims, not completion of every development feature. Select one UAV collection–processing graph and one supported DI partition first. Keep pipeline/tensor/hybrid generality, LLM streaming, and sophisticated recovery only where the implementation and evidence can support them. A negative performance result is an acceptable research finding when its conditions and costs are explained.

Table 6: Revised internal targets; defense is conditional on evidence and committee approval.

Period	Deliverable / decision gate
September 2026	Agree on three RQs and bounded claims; freeze supported application graphs and implementation/evidence baseline; resolve correctness blockers
October 2026	Execute registered correctness and matched comparison cases; reduce optional scope if an end-to-end gate remains open
November 2026	Freeze accepted results and provenance; write RQ answers, costs, and limitations; keep missing results visibly unverified
December 2026	Deliver an integrated complete draft to the advisor; no new optional feature branch in the research argument
January 2027	Advisor/committee review, revisions, defense readiness decision, and confirmation of current university submission/notice requirements
February 2027	Target defense only after required reviews and evidence closure; allow time for substantive corrections
Spring 2027	Complete approved corrections and final submission by the applicable Graduate School deadlines; confirm exact dates separately

The February 2027 defense is a target, not a statement of readiness or an institutional deadline. CS guidance calls for an advisor-approved complete dissertation to reach committee members two weeks before the defense [21]. The Graduate School currently lists March 26, 2027 for the defended final ProQuest submission in Spring 2027 [22]; confirm current instructions before filing. If core end-to-end evidence is still missing at the January review, the committee must reconsider scope or timing. Final-defense slides must replace planned tests with actual

answers; changing a label from “planned” to “completed” cannot satisfy that gate.

6.2 Limits That Must Remain Visible

ABE adds authority, computation, and rekeying costs; disconnected revocation has propagation limits. Candidate ACKs do not prove future capacity or honest reporting. Cache availability does not guarantee task completion. Local tests do not qualify distributed failure behavior. UAV safety and perception accuracy need evidence beyond service-message delivery. Hardware/model-specific findings cannot establish unlimited framework generality. These are conditions for interpreting the research, not reasons to omit an unfavorable result.

Chapter 7

Conclusion

NDNSF's proposed research contribution is a service-level protocol and evidence-backed design rationale, not a new encryption primitive or the first use of named computation. RQ1 asks when protected discovery and reuse of ABE capability are preferable to complete alternatives. RQ2 asks how runtime assignments become local rules for accepting named dependencies. RQ3 asks where the resulting costs buy useful completion, latency, or execution-efficiency benefits. Existing campaigns establish limited invocation findings; the collaboration and cross-application claims still require their declared end-to-end evidence. This separation provides a concrete route from the present proposal to a defensible dissertation.

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